Cø / N* Instructor-Controlled Student Reveal Sheets

Key-free staged evidence cards - do not distribute as a complete student pack

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Reveal-Sheet Use Guide

These sheets contain no answer keys, but they include evidence that must remain hidden until a team reaches the stated commitment point. Instructors distribute only the assigned sheet or page, never this complete file. All cases and values are synthetic and for teaching only.

Reveal Sheet - Session 3 Audit Card 1

Internal perturbations had twice the calibrated budget of external perturbations.

Reveal Sheet - Session 3 Audit Card 2

The preferred candidate ranking reverses under a defensible per-variable normalization and dimension-matched null.

Reveal Sheet - Session 3 Audit Card 3

The sampled external interventions have almost no support in the declared target regime.

Reveal Sheet - Session 4 Complication Card 1

A second credible N_1 surrogate remains at $.69[.63,.75]$.

Reveal Sheet - Session 4 Complication Card 2

Candidate reliability of the frozen P_{Θ} referent falls to $R_C = .71$, and the separate cross-condition mapping check fails.

Reveal Sheet - Session 4 Complication Card 3

N_2 falls to $.57[.51,.63]$ with paired change $\Delta N_2=[-.23,-.11]$.

Reveal Sheet - Session 4 Complication Card 4

The anchor decline occurs only on report trials.

Reveal Sheet - Session 4 Complication Card 5

Pathway-specific rescue restores N_1 and the anchors while the retained gates remain equivalent.

Reveal Sheet - Session 6 Team A

- **Early interruption, 55-85 ms:** fidelity $.40[.34, .46]$; $V = .79[.73, .85]$; $N_1 = .51[.45, .57]$; $N_2 = .48[.42, .54]$; $N_3 = .31[.25, .37]$; report accuracy $.33[.27, .39]$. Paired changes are $\Delta V = [-.04, .02]$, $\Delta N_1 = [-.29, -.15]$, and $\Delta N_2 = [-.28, -.16]$; $R_C = .88$ and mapping passes.
- **Target return-path cut, 145-180 ms:** fidelity $.82[.77, .87]$; $V = .79[.73, .85]$; $N_1 = .71[.65, .77]$; $N_2 = .68[.62, .74]$; $N_3 = .39[.33, .45]$; report accuracy $.58[.51, .65]$. Paired changes are $\Delta V = [-.04, .02]$, $\Delta N_1 = [-.06, .02]$, and $\Delta N_2 = [-.06, .02]$; $R_C = .90$ and mapping passes.

Classify both conditions and state which causal comparison these two rows can and cannot identify.

Reveal Sheet - Session 6 Team B

- **Target return-path cut, 145-180 ms:** fidelity $.82[.77, .87]$; $V = .79[.73, .85]$; $N_1 = .71[.65, .77]$; $N_2 = .68[.62, .74]$; $N_3 = .39[.33, .45]$; report accuracy $.58[.51, .65]$. Paired changes are $\Delta V = [-.04, .02]$, $\Delta N_1 = [-.06, .02]$, and $\Delta N_2 = [-.06, .02]$; $R_C = .90$ and mapping passes.
- **Late report-path cut, 360-390 ms:** fidelity $.83[.78, .88]$; $V = .80[.74, .86]$; $N_1 = .72[.66, .78]$; $N_2 = .69[.63, .75]$; $N_3 = .66[.60, .72]$; report accuracy $.29[.23, .35]$. Paired changes are $\Delta V = [-.03, .03]$, $\Delta N_1 = [-.05, .03]$, and $\Delta N_2 = [-.05, .03]$; $R_C = .91$ and mapping passes. The paired change in report accuracy is $[-.65, -.49]$.

Classify both conditions and state which causal comparison these two rows can and cannot identify.

Reveal Sheet - Session 6 Team C

- **Target return-path cut, 145-180 ms:** fidelity $.82[.77,.87]$; $V=.79[.73,.85]$; $N_1=.71[.65,.77]$; $N_2=.68[.62,.74]$; $N_3=.39[.33,.45]$; report accuracy $.58[.51,.65]$. Paired changes are $\Delta V=[-.04,.02]$, $\Delta N_1=[-.06,.02]$, and $\Delta N_2=[-.06,.02]$; $R_C=.90$ and mapping passes.
- **Sham, 145-180 ms:** fidelity $.83[.78,.88]$; $V=.80[.74,.86]$; $N_1=.72[.66,.78]$; $N_2=.69[.63,.75]$; $N_3=.67[.61,.73]$; report accuracy $.84[.78,.90]$. Paired changes are $\Delta V=[-.03,.03]$, $\Delta N_1=[-.05,.03]$, and $\Delta N_2=[-.05,.03]$; $R_C=.91$ and mapping passes.

Classify both conditions and state which causal comparison these two rows can and cannot identify.

Reveal Sheet - Session 6 Team D

- **Target return-path cut, 145-180 ms:** fidelity $.82[.77,.87]$; $V=.79[.73,.85]$; $N_1=.71[.65,.77]$; $N_2=.68[.62,.74]$; $N_3=.39[.33,.45]$; report accuracy $.58[.51,.65]$. Paired changes are $\Delta V=[-.04,.02]$, $\Delta N_1=[-.06,.02]$, and $\Delta N_2=[-.06,.02]$; $R_C=.90$ and mapping passes.
- **Pathway rescue after target cut, 145-180 ms:** fidelity $.82[.77,.87]$; $V=.79[.73,.85]$; $N_1=.71[.65,.77]$; $N_2=.68[.62,.74]$; $N_3=.65[.59,.71]$; report accuracy $.82[.76,.88]$. Paired changes are $\Delta V=[-.04,.02]$, $\Delta N_1=[-.06,.02]$, and $\Delta N_2=[-.06,.02]$; $R_C=.90$ and mapping passes.

Classify both conditions and state which causal comparison these two rows can and cannot identify.

Reveal Sheet - Session 6 Complete Synthesis

Release only after all teams record their partial-evidence limits.

Synthetic condition	Interval	Fidelity	V	N_1	N_2	N_3	Report accuracy
Early interruption	55-85 ms	.40[.34,.46]	.79[.73,.85]	.51[.45,.57]	.48[.42,.54]	.31[.25,.37]	.33[.27,.39]
Target return-path cut	145-180 ms	.82[.77,.87]	.79[.73,.85]	.71[.65,.77]	.68[.62,.74]	.39[.33,.45]	.58[.51,.65]
Late report-path cut	360-390 ms	.83[.78,.88]	.80[.74,.86]	.72[.66,.78]	.69[.63,.75]	.66[.60,.72]	.29[.23,.35]
Sham	145-180 ms	.83[.78,.88]	.80[.74,.86]	.72[.66,.78]	.69[.63,.75]	.67[.61,.73]	.84[.78,.90]
Pathway rescue after target cut	145-180 ms	.82[.77,.87]	.79[.73,.85]	.71[.65,.77]	.68[.62,.74]	.65[.59,.71]	.82[.76,.88]

Paired-change intervals for the retained components are:

Synthetic condition	ΔV	ΔN_1	ΔN_2	R_C / mapping status
Early interruption	$[-.04,.02]$	$[-.29,-.15]$	$[-.28,-.16]$	$.88$ / pass
Target return-path cut	$[-.04,.02]$	$[-.06,.02]$	$[-.06,.02]$	$.90$ / pass
Late report-path cut	$[-.03,.03]$	$[-.05,.03]$	$[-.05,.03]$	$.91$ / pass
Sham	$[-.03,.03]$	$[-.05,.03]$	$[-.05,.03]$	$.91$ / pass
Pathway rescue after target cut	$[-.04,.02]$	$[-.06,.02]$	$[-.06,.02]$	$.90$ / pass

For the late report-path cut, the paired change in report accuracy is $[-.65,-.49]$; this is a stipulated synthetic teaching interval, not an empirical estimate.

Reveal Sheet - Session 7 Variant A

Applicability and quality pass. The evidence score is $.78\,[.73, .83]$. Two independent surrogates agree; the P_{Θ} referent is stable and the positive bearer is resolved.

Reveal Sheet - Session 7 Variant B

Applicability and quality pass. The evidence score is $.55\,[.47, .63]$. There is no credible conflict.

Reveal Sheet - Session 7 Variant C

Applicability and quality pass. The aggregate score is $.75$, but EEG gives $.79\,[.74, .84]$ and imaging gives $.43\,[.37, .49]$; both surrogates are credible.

Reveal Sheet - Session 7 Variant D

The task is validated for this injury class and signal quality passes. The score is $.18\,[.12, .24]$, but severe hearing loss was not resolved and an auditory command task was used.

Reveal Sheet - Session 7 Variant E

The new neural measure has no validation outside healthy adults, although the recording is technically clean. Applied to this severe-injury patient, its score is $.76\,[.70, .82]$; the lower bound exactly equals τ_+.

Reveal Sheet - Session 7 Variant F

Applicability passes with sensitivity $.93$ in this injury class, and quality passes. The score is $.19[.13,.25]$. Detection opportunity, sensory route, stable P_{Θ} referent, and auxiliaries pass.

Reveal Sheet - Session 8 Extension Rows 9-12

Release after the core criterion-level verdict is committed.

Row	Weight	V	N_1	N_2	N_3	R_*	R_{-2}	Additional disclosed fact
9	.05	.75-.85	.70-.80	.37-.47	.68-.78	.14	.48	anchor reuses the N_2 recipient score
10	.05	.72-.82	.73-.83	.39-.49	.70-.80	.22	.45	frozen-referent R_C=.67 and cross-condition mapping fails
11	.05	.74-.84	.72-.82	.41-.51	.69-.79	.39	.38	applicability and measurement quality pass
12	.10	.76-.86	.75-.85	.72-.82	.71-.81	.17	.18	applicability and measurement quality pass

For the full valid diagnostic set after extension, use the supplied synthetic bootstrap interval $\Delta_2 = R_{-2} - R_* : [.08, .21]$. Freeze the revised criterion-level claim before requesting a mechanism dossier.

Reveal Sheet - Session 8 Mechanism Dossiers

Two synthetic mechanism dossiers accompany the table:

- **Dossier M1:** a route-specific interruption lowers N_2 below threshold with $\Delta N_2 = [-.31, -.19]$; frozen-referent $R_C = .90$ and the separate cross-condition mapping check passes; paired-change intervals for V, N_1, N_3 are contained in $[-.08, .08]$; content fidelity, route mediation, matched control, and pathway rescue pass. The full candidate lattice and alternative candidates' post-intervention profiles are not supplied.
- **Dossier M2:** a global suppression lowers N_2, also lowers V, N_1, N_3 outside their equivalence margins, lowers frozen-referent R_C to $.61$, and fails cross-condition mapping; generalized anchor decline follows.

Use the risk table for criterion-level incremental value. Use Dossiers M1 and M2 to decide whether either intervention supports a selective mechanism claim. Do not convert a measurement ablation into a physical-necessity conclusion. Freeze this second verdict before the instructor releases one midpoint audit card.

Reveal Sheet - Session 8 Audit Card 1 / Off-Manifold Enrichment

The two largest diagnostic rows came from an engineered preparation absent from the declared target population; their printed weights cannot be used as population coverage.

Reveal Sheet - Session 8 Audit Card 2 / Post-Anchor Selection

Diagnostic rows were selected after the analyst inspected the supposedly held-out anchor risks.

Reveal Sheet - Session 8 Audit Card 3 / Referent-Mapping Failure

Under Dossier M1, frozen-referent R_C is newly found to be $.64$ and the separate cross-condition mapping check fails.

Reveal Sheet - Session 8 Audit Card 4 / Dual-Use Outcome

The supposedly independent anchor vector includes the same recipient-route score used to construct N_2.

Reveal Sheet - Session 9 Variant A

Baseline phenomenal distances for B: to R, O, G	Held-out rank accuracy	Intervention prediction for R-O	Observed R-O after intervention	Validity
2.5, 1.5, 1.1	.92	contracts from 1.0 to .50	$.55[.48, .62]$	attention, retinal input, task, and response controls pass

Reveal Sheet - Session 9 Variant B

Baseline phenomenal distances for B: to R, O, G	Held-out rank accuracy	Intervention prediction for R-O	Observed R-O after intervention	Validity
2.6, 1.6, 1.0	.94	contracts from 1.0 to .50	$1.02\,[.94, 1.10]$	attention, retinal input, task, and response controls pass; anchors were not used to tune intervention

Reveal Sheet - Session 9 Symmetry Extension

Three unlabeled qualities form an equilateral triangle. List the vertex permutations left possible by distance alone. Then add a validated negative-valence label to one vertex and state which symmetries remain.

Reveal Sheet - Session 10 Profile A

P_{Θ} is stable at $.91$; no all-pass proper subsystem exists; B^{+} is resolved. $V = .80[.74, .86]$, $N_1 = .72[.66, .78]$, $N_2 = .70[.64, .76]$, and $N_3 = .58[.52, .64]$. Validation anchors were constructed independently.

Reveal Sheet - Session 10 Profile B

P_{Θ} autonomy / mapping stability is $.62$. $V = .79$, $N_1 = .71$, $N_2 = .69$, and $N_3 = .57$. The result reverses across two reasonable referents.

Reveal Sheet - Session 10 Profile C

P_{Θ} is stable at $.90$. $V = .81$, $N_1 = .73$, $N_2 = .38[.31, .45]$, and $N_3 = .60$. Route, source, and retained-gate checks pass; the complete candidate lattice is not supplied.

Reveal Sheet - Session 10 Profile D

P_{Θ} is stable at $.89$. $V = .78$, $N_1 = .49\,[.42, .56]$, $N_2 = .67$, and $N_3 = .37\,[.30, .44]$. The intervention intended for N_3 also reduces N_1.

Reveal Sheet - Session 10 Profile E

P_{Θ} is stable at $.92$; no all-pass proper subsystem exists; B^{+} is resolved. $V = .82$, $N_1 = .75$, $N_2 = .72$, and $N_3 = .61$. The supplied “ground truth” labels were generated by thresholding these same component scores.

Reveal Sheet - Session 11 Stage 2 / Round 3

Component and validity card. During one high-arousal window, an imagery classifier exceeds its healthy-control threshold on two artifact-free, cross-validated runs out of ten. Its frozen task-level positive rule is at least two qualifying runs with familywise false-positive probability below $.05$. The classifier has good specificity but modest sensitivity in this injury class, and hearing demand was satisfied for this high-arousal window. $N_1 = .59[.51, .67]$ against threshold $.60$ and $N_3 = .48[.40, .56]$ against threshold $.50$, so both intervals straddle. A content-specific N_2 perturbation was not performed. During a later low-arousal window, an EEG assessment is negative but contaminated by muscle artifact; hearing was not reverified for that later window.

Before requesting Stage 3, freeze the Stage-2 scientific outcomes, the strongest permitted conclusion, and one forbidden inference for each target. Do not infer the contents of the action card.

Reveal Sheet - Session 11 Stage 3 / Round 4

Action card. A low-burden visual communication trial is available. Missing genuine command following carries 20 loss units; undertaking the trial carries 1 loss unit regardless of outcome. On these simplified losses, the trial is favored when the decision maker's probability of genuine command following exceeds .05.

Reveal Sheet - Session 11 Longitudinal Extension

Reclassify these three synthetic windows without changing the target:

Window	Evidence
Day 2	no response; hearing unverified; artifact heavy; viability unstable
Day 9	hearing verified; P_{Θ} referent stable; N_2 EEG positive and imaging negative; both credible
Day 21	repeated imagery modulation; V, N_1, N_2, N_3 pass for the command-related content; no all-pass proper subsystem remains and the minimal bearer is resolved; no motor output

Reveal Sheet - Session 12 Dossier A / Octopus

PRI is $.72[.67,.77]$. Perturbational differentiation appears similar although sensor layout differs. Distributed arm-brain effects are demonstrated under route interruption. No target-domain rule-in/rule-out study exists and the source threshold was copied. The mapped organism P_{Θ} referent is stable for the trial. The theory’s other required gates are stipulated to pass for this specified trial.

Reveal Sheet - Session 12 Dossier B / Six-Month Infant

PRI is $.69[.66,.72]$. Homologous signal genealogy and artifact controls pass. Conserved thalamocortical causal role and matched perturbation response pass. Preregistered developmental calibration gives sensitivity $.87$, specificity $.84$, and threshold $.64$. Boundary and sleep-wake regime are stable. The theory's other required gates are stipulated to pass for this specified trial.

Reveal Sheet - Session 12 Dossier C / Organoid-Controller Hybrid

PRI is $.76[.70,.82]$. The same analysis code produces a high value, but a controller pulse imposes most of the differentiated response. No phenomenal or role calibration exists in hybrids. Tissue-only and hybrid verdicts reverse, and the controller regime changed during testing. Because candidate and regime are unstable, the theory's other gates cannot be interpreted.

Reveal Sheet - Session 12 Dossier D / Tool-Using Artificial Agent

PRI is $.81[.75,.87]$. The index detects a complex perturbational response. State reset demonstrates some internal role, but scheduler and human selection dominate long-horizon behavior. No artificial-domain calibration exists and the biological source threshold was copied. A short-horizon candidate is plausible; the long-horizon boundary is indeterminate. V, N_1, N_2, N_3 and the long-horizon boundary remain unassessed.

Reveal Sheet - Session 13 Profile A / IIT Contrast

An IIT 4.0 analysis identifies one exclusion-qualified complex with specified intrinsic cause-effect structure. The Cø / N* candidate is viable and N_1 passes, while N_2 and N_3 fail under licensed tests. Report is absent.

Reveal Sheet - Session 13 Profile B / GNWT Contrast

A content undergoes a source-qualified workspace ignition and becomes globally available to multiple processors. N_2 passes, while licensed N_1 and N_3 tests fail. Flexible task use occurs.

Reveal Sheet - Session 13 Profile C / RPT Contrast

Content-specific local sensory recurrence passes the Lamme-version criterion; N_1 and N_3 pass, while licensed reduced-family tests make N_2 fail. Executive and report routes are absent.

Reveal Sheet - Session 13 Profile D / Higher-Order Contrast

A first-order perceptual state has the source-specified higher-order representation. V and N_1 pass, while N_2 and N_3 fail under licensed tests.

Reveal Sheet - Session 13 Profile E / Full-Bundle Challenge

S is a stable P_{Θ} referent and V, N_1, N_2, N_3 pass; no all-pass proper subsystem exists, the positive-bearer relation is resolved, and $S \in \hat{B}^{+}_{min,\Theta}(c)$. A separately validated anchor vector falls in a strong rule-out region; every source, independence, boundary, and validity license passes.

Reveal Sheet - Session 14 Result Card 1

Model-risk difference is $.11\,[.06, .16]$, but diagnostic coverage is $.06$.

Reveal Sheet - Session 14 Result Card 2

Risk difference is $.01[-.02,.04]$, coverage is $.28$, retained components and sensitivity checks pass, model flexibility and transport are matched, no compensatory retuning occurs, and the complete interval is contained in the frozen closed equivalence region.

Reveal Sheet - Session 14 Result Card 3

Apparent risk difference is $.14\,[.08, .20]$, but frozen-referent R_C falls from $.91$ to $.62$ under intervention and the separate cross-condition mapping check fails.

Reveal Sheet - Session 14 Result Card 4

Anchor family A gives $.12[.08,.16]$; equally credible family B gives $-.07[-.11,-.03]$; the shared-channel audit passes.

Reveal Sheet - Session 14 Result Card 5

The source-domain effect replicates, but target-domain calibration and causal-role mapping fail.